Frequently Listed, Rarely Examined: A Screened Bibliometric Map of Warming and Cooling Food Claims in Japan

Running title: What predicts research attention to warm/cool food claims in Japan

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Abstract

Background: Japanese *onkatsu* ("warming care") rests on a widely shared belief that particular foods warm or cool the body. Prior reviews find the support heterogeneous rather than absent, but none has asked whether the most widely listed claims are the examined ones.

Methods: Cross-sectional bibliometric study. Axis A counts how many of 15 Japanese lay-facing sources (frozen in advance; nine form the primary frame) assign each food a warming or cooling direction, from a ledger of every candidate span. Axis B counts PubMed hits co-occurring with 22 thermal-effect terms, then screens every hit, the raw count measuring co-occurrence, not the construct; survivors were sub-labelled by whether that was the study's own question. Two language models of adjacent tiers from one vendor labelled every set independently (18,730 candidates, Cohen's κ = 0.8998; 12,733 records, 0.8068; 298 sub-labels, 0.7804), the author adjudicating.

Results: Screening cut 98% of the raw signal: 11,714 hits across 146 foods became 284 on-construct studies; 96 foods (65.8%) had none. Adjusting for total PubMed volume (L1) — the opportunity to be studied — coverage showed no association (odds ratio [OR] 1.062, 95% confidence interval [CI] 0.876–1.289, $p = 0.539$), while volume did (OR 2.083 per unit of $\log(L1 + 1)$, 1.568–2.767, $p < 0.001$). It held under three other forms of that adjustment (1.07–1.13) and under the narrowest reading, the 171 studies asking that question themselves (1.086, 0.890–1.324). Unadjusted, coverage was associated with study presence (OR 1.303, 1.101–1.541, $p = 0.002$). Chicken, warming in six sources: 702 hits, five studies, one asking it.

Conclusions: With literature volume held fixed, no association was detected between coverage and whether a food had been studied with a qualifying thermal outcome, under either reading of "studied". Compounded over one-to-nine sources the interval spans 0.35–7.62-fold, so a substantial effect is not excluded. For two-thirds of these foods, the search retrieved none.

Keywords: attention gap; bibliometrics; folk nutrition; warming and cooling foods; thermoregulation; abstract screening; Japan

Introduction

The idea that individual foods “warm” or “cool” the body is a durable feature of everyday health culture in Japan, where it underpins the popular practice of *onkatsu* (温活, “warming care”) and lay reasoning about *hiesho* (冷え症, habitual cold sensitivity) — a complaint recorded distinctly enough in Japanese public health statistics that a nationwide survey of 26,131 working-age respondents estimates its prevalence separately by sex (Miyoshi et al. 2026). Corporate wellness media, food manufacturers, and individual practitioners routinely publish lists that sort ginger, carrot, and burdock as “warming” and cucumber, tomato, and coffee as “cooling,” and these attributions circulate widely enough to shape ordinary purchasing and seasonal eating advice. That the belief is culturally established, rather than niche, is visible both in the volume of lay-facing sources that maintain such lists and in the fact that Japanese epidemiology has taken the classification seriously: Nagata et al. (2017), analyzing the Takayama cohort (n = 28,356), applied four independent warm/cool food-classification lists to real dietary data. The premise of the belief is therefore neither obscure nor recent.

Crucially, the scientific literature on warm/cool food theory is not empty, and this study does not claim that it is. Ormsby (2021), in a scoping review of the nutritional evidence, characterized the support as “heterogeneous and of mixed quality” while identifying partial mechanistic correlates — heating foods associated with higher caloric density, sympathetic activation, and vasodilation, and cooling foods with higher water and fiber content and anti-inflammatory processes. Namiranian et al. (2021), a companion review in the same volume, surveyed the physiological basis of hot/cold theory across Persian medicine, traditional Chinese medicine, and Ayurveda. Zhou & Xu (2021) integrated candidate molecular mechanisms (for example, differential nuclear factor kappa B [NF-κB] and mitogen-activated protein kinase [MAPK] signaling) linking the cold/hot nature of foods to biological effects — while explicitly noting that research applying these mechanisms to foods is scarce relative to research on Chinese medicines. Any framing of the field as an “evidential desert” is thus untenable; the appropriate description is fragmentary, unintegrated, and unevenly attended.

At the level of individual foods, the available evidence is not only fragmentary but sometimes contradictory or opposite to belief. Ginger, the archetypal “warming” food, showed a thermic-effect difference at $p = 0.049$ in one small randomized crossover pilot — though total resting energy expenditure and the thermic-effect area under the curve (AUC) were both null in that same trial (Mansour et al. 2012, n = 10 overweight men) — while a larger, double-blind crossover trial in a different population reported no increase at all (Fagundes et al. 2021, n = 20 normal-weight women). Coffee, popularly classified as “cooling” (or “yin”) in Japan, is dominated by caffeine, whose best-powered synthesis under thermal stress — a meta-analysis pooling n = 30 caffeine effects — found that caffeine significantly *raises* peak core temperature (Hedges’ $g = 0.44$; Peel et al. 2025). The physiological direction most relevant to coffee's principal active compound therefore runs opposite to the folk attribution, even after allowing for the distinct measurement context (heat-exposure performance settings rather than everyday thermal sensation).

Methodologically, the practice of mapping a hot–cold belief system bibliometrically has direct precedent. García-Hernández et al. (2023) assembled and classified 101 academic publications spanning roughly a century to characterize Mexico's hot–cold system by research approach, depth, and conceptual domain. Nagata et al. (2017) demonstrated, within Japan, that the four extant warm/cool classification lists disagree with one another when applied to cohort data — an internal inconsistency that is itself a symptom of the field's uneven development. These works establish that belief systems of this kind can be quantified and that the classification schemes underlying them are neither unified nor exhaustively validated.

What remains unquantified is the *asymmetry* between the two facts above: that certain foods are very widely presented as warming or cooling, and that the direct scientific attention paid to those specific claims varies enormously. Ormsby (2021) asked whether mechanistic support exists but did not measure how widely each food is claimed, nor compare research volume across foods; reviews of the adjacent clinical concept of cold hypersensitivity (Jin et al. 2025, 65 studies) consistently take traditional-medicine interventions, not lay food beliefs, as their subject. We therefore ask a descriptive question: across the foods that Japanese lay sources classify as warming or cooling, is the breadth of that lay coverage related to whether the food has been studied with an outcome that could speak to the thermal claim — and, if there is any apparent relationship, does it survive adjustment for how much the food is studied for any reason at all?

Answering that question turns out to depend on a measurement problem that a keyword count cannot solve, and addressing it is the methodological contribution of this paper. Counting how often a food name co-occurs with thermal vocabulary in PubMed does not count studies of that food's thermal effect in people: the co-occurrence set is dominated by poultry and livestock heat-stress research, and it misses the nutrition trials that report the thermic effect of food. We therefore screened the entire co-occurrence set to the intended construct before analysing it, using two independent language models with author adjudication — a design with recent published precedent in evidence synthesis (Guo et al. 2024; Hilkenmeier et al. 2026).

Methods

Design

This is a descriptive, cross-sectional bibliometric study. It maps two independently constructed axes against each other for a fixed set of foods and reports their association; it does not estimate a causal effect of lay coverage on research or vice versa. Because the study was not registered on any protocol registry, we describe fixed-in-advance choices as *planned* rather than pre-specified.

Axis A: lay-source coverage

Axis A quantifies how many independent Japanese lay-facing sources assign each food a warming or cooling direction, following the bibliographic-coding approach of García-Hernández et al. (2023). We use "coverage" rather than "belief" throughout: the measure counts sources that publish an attribution, and is a proxy for — not a measurement of — how widely the attribution is actually held.

How the frame was assembled. Candidate sources were located by Japanese-language web search in two passes. The first swept general queries for the belief itself — among them 「体を温める食べ物 冷やす食べ物 一覧 漢方」 ("list of foods that warm or cool the body, kampo"), 「温活 食材リスト 温める冷やす 分類」 ("onkatsu ingredient list, warming/cooling classification"), 「五性 温 熱 涼 寒 食材 一覧」 ("five-nature ingredient list"), and 「冷え性 食べ物 温める 冷やす 一覧」 ("hiesho food list, warming and cooling"). The second ran domain-restricted queries over the sectors that publish such lists: kampo and pharmaceutical manufacturers, health-food companies, drugstore chains, food and beverage makers, medicinal-food (*yakuzen*) associations, and the web editions of women's and lifestyle magazines. Two independent existence scans then verified every candidate by fetching the page — publishing about warming care in general was not sufficient; the page had to assign individual foods a direction — and by retrieving its robots.txt. Independence was judged on operator and on content, so aggregator pages republishing another site's list were dropped. The frame is therefore a purposive sample of what was reachable on the Japanese-language web in August 2026, assembled before any Axis B count existed; it is not a probability sample of Japanese lay media, and it is not weighted by readership.

The sampling frame was frozen in advance: 15 sources verified to be reachable over HTTP, to present a concrete per-food warm/cool list, and to permit fetching per robots.txt. Nine corporate- or association-operated sources form the **primary (Tier 1) frame**; six individual-expert blogs form a **Tier 2 sensitivity frame**, which the coding protocol reserves for testing whether conclusions are stable. Sources sharing a single operator were counted once. Two frame corrections were made during fetching and before any coding of results — one source lacking a concrete per-food list was excluded, and one source's URL was corrected to its per-food list page — with both decisions driven by page content rather than by any result.

Each source page was fetched once, on 2026-08-03 (one source re-fetched 2026-08-04 at its corrected URL), and stored locally; the per-source access dates are published in `data/sources.csv`. Every food mentioned was then coded for its direction (warming, cooling, or neutral), with the verbatim supporting quotation recorded on each row. Coding rules were documented before coding: sources using traditional five-nature or yin–yang vocabulary were mapped to warming (温性/熱性/陽性), cooling (涼性/寒性/陰性), or neutral (平性); serving temperature (e.g. "cold drinks") is not a food's nature and was not coded as one. For each food, Axis A records the number of independent sources assigning a direction (`n_sources`) and the majority direction; a consensus ratio is also computed and published with the data, but no analysis here uses it.

Who coded, and what became of the first pass. All coding was performed by large language models operating under the author's direction; there were no human raters. The frame was coded twice in the study's history, and both passes are published. The first, in August 2026, read each source and recorded the foods it credited with a direction; it is retained unchanged as `data/claims_frozen.csv`, and its role is now that of a reference set rather than of the analysed data, since the completeness checks below are measured against it and regenerating it would make them self-referential. What that pass could not show was what had been read and set aside. It kept 649 retained rows and no record of any exclusion, so a reader could confirm that each row was real but not that nothing was missing — two different claims, and only the first was supported. The second pass was built to support both.

A ledger over every candidate. The Axis B screening publishes 298 includes alongside 12,435 reasoned exclusions, which lets a reader check the exclusions rather than take them on trust. Axis A is now constructed the same way: every candidate the sources present receives an include-or-exclude decision with a reason code, and the whole ledger is published as `data/claims_ledger.csv`. The decision rules are documented in `data/coding_protocol.md` §9, written before this pass ran, and each of its exclusion codes is derived from a rule the protocol already stated or from an adjudication it already recorded, rather than invented for the ledger.

Enumerating candidates. Candidates are enumerated from the structure the sources use to present foods — delimiter-separated lists, `category：item` lines, one-item-per-line blocks under a thermal heading, and running text — and from no food dictionary. Searching the sources with vocabulary drawn from the existing coding would be self-referential in the same way as regenerating the reference set: a food no source had ever been credited with could not be found that way, which is precisely the failure the ledger exists to detect. A candidate is a span rather than a resolved food name, because Japanese running text offers no reliable token boundary for foods written in kana — じゃがいも ends in the particle も, たけのこ contains の — so the extractor emits overlapping granularities and leaves the naming to the coder, which is what the direction rules already ask of one. A coarse span is still judgeable; a food that never became a span is not. This yielded **18,730 distinct (source, span) candidates** from 48,639 occurrences, drawn from 3,323 of the frame's 3,852 body-text lines.

How complete the enumeration is. Three checks, and the last two exist because the first cannot stand alone. Measured against the frozen coding, **every one of its 649 rows is surfaced** as a candidate — covered-recall 1.0000, and 1.0000 within Tier 1 alone, with 94.0% recovered as an exact token rather than inside a coarser span. That is a necessary condition and not a sufficient one, because recall against coded rows cannot speak for a food no coder ever recorded. Two checks that read no coded data at all are therefore reported beside it. Of the **3,852 body-text lines** in the frame, **1,136 carry thermal vocabulary, and every one of those yields at least one candidate**; since the protocol forbids coding a direction the source does not state, a line making an attribution must carry that vocabulary, so no attributing line goes unexamined. Separately, each of the 15 sources was read end to end by an agent asked only to list the foods that source gives a direction, blind to both the extraction and the frozen coding: of the **810 foods** those reads returned, all 810 appear among the candidates. The reads are published verbatim in `data/independent_read/`.

Two coders, and what the agreement statistic measures here. Every candidate was labelled independently by two large language models of different capability tiers from one vendor's line — Claude Sonnet 5 and Claude Opus 5 — which is the instrument the Axis B screening uses, so both axes are judged by the same one. Each coder received the protocol and its assigned candidates only; the other coder's labels, the frozen coding and the reconciliation code were withheld. As on Axis B, this pairing measures within-lineage consistency rather than independence, and we claim no more for it. What changes is the quantity the statistic is computed on. The κ of 1.000 reported for the first pass was a *direction* statistic over items both coders had already extracted, and extraction was where that pass's disagreement actually lived — the extraction overlap in its second round was a Jaccard index of 0.54, nothing near unity. Because both coders now judge the same enumerated candidate set, extraction is no longer a source of divergence, and κ is computed over the include/exclude decision across every candidate: **κ = 0.8998 on all 18,730** (observed agreement 0.9916), with 157 divergences.

The author adjudicated **208 candidates** against the source text. Two of the three routes into that set are the ones Axis B uses — every divergence, and every candidate a coder flagged as uncertain — and the third is one a divergence list cannot produce: candidates both coders labelled identically where the protocol had no rule, which coders were instructed to report rather than resolve on their own. Two independent readers of an unruled case tend to fall to the same conservative default, so agreement there is not evidence that the label is right, and a design that watches only κ and the divergence list loses exactly as much as the rules leave unstated. Every ruling is filed with its rationale in `data/ledger_rulings.csv` and flagged in the ledger, so no label changes silently.

A human audit of the ledger. Every judgment above was made by a language model, so the agreement statistic bounds consistency within one vendor's lineage rather than accuracy — a general weakness that becomes a specific one here, because measurement error in `n_sources` produces exactly the non-finding this paper reports. Part of the ledger was therefore read back against the sources by the author, under a procedure written and published before it was performed (`data/coding_protocol.md` §10; the sampling seed is published and the empty sheets were committed before any row was read). It has two halves. For **precision**, a simple random sample of **60 of the 497 Tier-1 claim rows** was read against its source and given one of five verdicts fixed in advance. The reading was done in a worksheet showing each row beside the archived copy of its page — the text the coders themselves were given — with the live page linked from every row. Section 10 specifies the live page, so this is a departure from it and is recorded as one: it can only push the precision figure up, because a page edited since the freeze would be unverifiable against the live copy and is verifiable against the archived one. No row was ruled unverifiable. For **completeness**, **two of the nine Tier-1 sources** — the longest by extracted body text, plus one drawn at random from the remaining eight — were read end to end, every food the source gives a direction listed without reference to the ledger, and that list then diffed against it. Completeness is the half that answers the objection: a missed claim lowers a food's `n_sources` and compresses the predictor's spread, whereas a spurious one mostly adds noise. Names the ledger does not hold under the surface form the reader wrote are adjudicated afterwards against the coding rules, each ruling published in `data/human_audit_rulings.csv` with its reason; the results are reported in the Limitations, at both the specified and the adjudicated count.

What was excluded, and why. The ledger records **898 includes and 17,832 exclusions**. The reason codes are `fragment` (10,199 — a string that is not a unit of the source, typically a token cut out of a longer name), `navigation` (5,666 — site furniture, menus, page and section titles), `not-food` (1,741), `no-direction` (173 — a food named on a line that predicates the direction of something else), `serving-temperature` (48 — chilled drinks and the like, which the protocol treats as a property of serving rather than of the food), `not-verbatim` (3) and `duplicate` (2). Category and umbrella labels are deliberately *not* excluded: 葉物野菜 ("leafy greens") or 香辛料 ("spices") is an attribution the source did make, and discarding it would misdescribe the source, so such a label is included and sub-labelled, with whether it can carry an Axis B measurement left to the query stage below.

What the coding yielded. The ledger's include rows generate `claims.csv` rather than being maintained beside it, so there is one record of the coding rather than two that can drift apart. Collapsing the several spans that can carry a single attribution — a food named in a list and again in the sentence beneath it — gives **824 rows over 301 distinct foods**, of which **218 are covered by at least one Tier-1 source**. Of those 218, **72 are labels that no single unambiguous query can represent**: 54 are classes the source names as a class, 16 are prepared dishes, and two are composite descriptions ("fatty meat", "mixed nuts"). These carry no Axis B measurement and are excluded from it, leaving **146 foods** in the primary analysis. A normative five-nature classification, against which lay attribution could be benchmarked, was planned but is deferred to future work because a citable primary source could not be secured within scope.

Axis B: on-construct research attention

Raw counts. Axis B begins from the National Center for Biotechnology Information (NCBI) PubMed E-utilities (no API key; queries rate-limited under 3 requests/s). All PubMed, CiNii, and OpenAlex queries were executed on 2026-08-06 to 2026-08-07; because bibliometric counts grow over time, this snapshot fixes the reported values. For each food we counted hits at two nested layers: **L1**, the food term alone, as total research volume; and **L2**, the food term co-occurring with any of **22 thermal-effect terms**. The vocabulary is set in one-to-one correspondence with the outcome classes the screening protocol accepts, and each term is searched verbatim as printed here: *body temperature*, *core temperature*, *skin temperature*, *tympanic temperature*, *axillary temperature*, *rectal temperature*, *oral temperature*, *thermogenesis*, *thermic effect*, *heat production*, *energy expenditure*, *metabolic rate*, *peripheral circulation*, *blood flow*, *microcirculation*, *thermoregulation*, *cold tolerance*, *cold exposure*, *thermal sensation*, *thermal comfort*, *cold sensitivity*, and *cold hypersensitivity*. The list is generated from `src/food_query_terms.py`, so a reader can check it against the query strings published with every count. An earlier version of this analysis used four terms (*thermogenesis*, *body temperature*, *peripheral circulation*, *thermoregulation*), which is narrower than the rule the screening then applies — and a query cannot measure the absence of evidence about an outcome class it never searches for. That narrower set demonstrably missed on-construct studies (see below), so it was replaced and all 175 queryable foods then in the frame were re-queried on 2026-08-08, and the 15 foods the Axis A rebuild later added were queried on the same 22 terms, bringing the queried universe to 190; both vocabularies and every count snapshot are published. Terms known to add noise were kept deliberately: plant "cold tolerance" pulls in agronomy records and "blood flow" pulls in veterinary ones, but both concepts sit inside the inclusion rule, and screening can remove noise whereas nothing downstream can recover a record the query never returned. Japan-specific foods with weak English coverage were given minimal English synonyms (for example, natto → "fermented soybean"); the inclusion criterion was held constant across foods so that counts are comparable. Queries are built deterministically from each food's key, and the raw count response for every query is published, so each reported count is pinned to its retrieval snapshot.

Why the raw count is not the measure. L2 counts keyword co-occurrence, not studies of the construct, and it fails in both directions. It over-counts heavily: `chicken` returned 702 hits dominated by avian thermoregulation and poultry heat-stress research, 566 of them (80.6%) animal studies. An analysis built on L2 would report the research attention paid to a keyword pattern, not to the belief. Screening addresses that failure. The opposite failure cannot be fixed downstream, because screening is a filter and L2′ is a subset of L2: it removes false positives but never recovers a study the query did not retrieve. This is why the effect vocabulary was rebuilt rather than left at four terms, and the case that forced it is concrete — Fagundes et al. (2021), a randomized double-blind crossover trial of ginger extract measuring thermic effect of food by indirect calorimetry alongside axillary temperature, was returned by none of the four terms. Under the 22-term query it is retrieved, and both coders independently labelled it `include`. Residual under-counting — evidence outside PubMed, or reported in vocabulary no term of ours captures — is carried into the Limitations as a floor on the measure.

Screening. We retrieved the title, abstract, journal, and publication types for every L2 hit and screened all **12,733 records across the 146 of the 190 queried foods that returned at least one hit** to a single construct: *a study of the thermal effect, on a human, of ingesting the food or its principal dietary constituent*. The include/exclude definition is documented in `data/screening_protocol.md` and fixed before coding. Include requires all of: human subjects in vivo; oral ingestion of the food, a food form of it, or its principal dietary constituent; a thermal-related physiological outcome (core, peripheral, skin, tympanic or axillary temperature; thermogenesis or thermic effect of food; energy expenditure reported in a thermogenic context; peripheral circulation; thermoregulation or cold tolerance; subjective thermal sensation); and attribution of the outcome to the food as the exposure of interest. **The construct is the study's outcome, not its motivation.** A trial that measures a qualifying thermal outcome after ingestion counts whether or not it set out to test a warm/cool belief, and most did not — the two chicken trials described below measure microvascular reactivity in an enriched-meat intervention. L2′ therefore counts studies that *could* speak to the claim rather than studies addressed to it, and "examined" is used in that sense throughout. **A sub-label separates the two readings.** Because those are different quantities and the second is the one a reader is likely to have in mind, every included record was additionally coded `claim-directed` or `incidental` under a section of the screening protocol written for the purpose (§8): `claim-directed` where the study's own framing — its title and its stated aim — names this food, a food form of it or its principal dietary constituent as what is given or varied *and* names a qualifying thermal outcome as what is measured; `incidental` where the record qualifies but the study was asking something else. The pass used the same two coders and the same reconciliation, agreement and adjudication machinery as the include/exclude pass, reaching κ = 0.7804 over the 298 included records (89.9% agreement) with 34 records adjudicated, 30 of them divergences, and settling at 178 claim-directed against 120 incidental. Section 8 was calibrated before the full pass on a 20-record canary stratified over its own boundary clauses, which raised κ from 0.588 to 0.886 and produced four revisions to the text; the canary labels were discarded and its records re-coded under the final wording. The pass cannot change an include/exclude decision, and its output is published as a separate ledger rather than written back into the screening ledger, so L2′ as reported is unchanged by it and it enters the analysis only as a sensitivity refit. Exclusions are coded by reason (`animal`, `livestock-heat`, `invitro`, `agri`, `name-only`, `no-thermal`, `mechanism-only`, `not-ingestion`, `species-mismatch`, and the two information-shortfall flags `uncertain-species` and `no-abstract`, which route a record to the author regardless of whether the coders agreed). Isolated constituents given at dietary doses (caffeine for coffee, capsaicin for chili pepper) are included and sub-labelled; a different species in the same family is excluded, so that *Aframomum* and *Kaempferia* studies do not inflate "ginger." No quality grading was performed — this is construct-validity screening, not risk-of-bias assessment. The 44 queried foods with zero L2 hits have L2′ = 0 by construction and required no screening; foods queried but not screened would be recorded as missing rather than zero, and there were none.

Three boundary questions arose during screening and were settled in the protocol before the affected records were judged, so that the same rule applied across every food. First, the *vascular bed*: "peripheral circulation" reaches cutaneous and limb perfusion — skin blood flow, forearm, calf, hand and foot flow, extremity rewarming — but not organ-specific perfusion measured for a non-thermal purpose (cerebral, coronary, renal, hepatic, retinal, ocular, sublingual, placental, splanchnic, or skeletal-muscle perfusion for nutrient delivery). Without that line, any vascular-function trial of any food becomes an include and L2′ fills with endothelial-function literature the folk claim never addressed. Second, *energy expenditure* qualifies where the paper frames it as thermogenesis, diet-induced thermogenesis, thermic effect of food or the food's thermogenic effect, and not where it appears as a covariate, as the denominator of an exercise energy balance, or as a body-composition or weight-loss endpoint. Third, *reactivity probes* — acetylcholine, sodium nitroprusside, methacholine, insulin-stimulated limb flow, flow-mediated dilatation, post-occlusive reactive hyperaemia — read in a hypertension or endothelial-function framing are endothelial endpoints rather than thermal ones, whereas **local thermal hyperaemia**, the skin's perfusion response to local heating, is thermal by construction and qualifies.

Screening reliability. Every record was labelled independently by two large language models of **different capability tiers from one vendor's line**: Claude Sonnet 4.6 with Claude Opus 4.7 for the 623 records of the five golden-set foods, and Claude Sonnet 5 with Claude Opus 5 for the 2,264 records of the remaining foods in the four-term set, for the 9,550 records the 22-term query added, and for the 296 records the Axis A rebuild brought in. Model versions differ across batches because screening ran over several working sessions; within each batch the two coders are contemporaneous. The recall-rebuild batch deliberately reuses the same two tiers as the batches carrying most of the existing judgments, so the widened record set is judged by the same instrument as the set it extends, and the 2,887 judgments made under the four-term query are reused verbatim rather than re-elicited — re-coding a record already judged would put two labels on one key. We state the pairing precisely because it bounds what the agreement statistic can mean: two models from one vendor and one release generation share pretraining data, tokenizer and alignment methodology, so their errors should be expected to correlate, and their agreement measures the **consistency** of that lineage rather than the convergence of independent judgments. This is a weaker design than the three-vendor arrangement of Hilkenmeier et al. (2026), and we do not claim its properties. Each agent received the protocol and its assigned records only — the other coder's labels, the golden set, and the reconciliation code were withheld. Cohen's κ was **0.8068 on all 12,733 co-coded records** (observed agreement 0.986), with 176 divergences; it was 0.811 on the 2,887 records of the four-term set alone, so widening the vocabulary did not destabilise how the protocol was applied. Both coders returned every record, so Axis B has none of the coverage differences that Axis A had to adjudicate, and κ's denominator here is the entire record set rather than an agreed subset of it. The author adjudicated **499 records** against title and abstract. The first pass settled 217: every divergence, plus every record either coder flagged as not stating the subject species or as having no abstract, since agreement reached on information a record does not contain is not evidence about that information. A further 156 rulings changed no label at all: `cream[tiab]` retrieves pharmaceutical and cosmetic creams alongside the dairy food, and both coders excluded those records but under different reason codes — one as a non-oral administration, the other as a name-only match — so the reason codes alone were reconciled, following the sense in which the frozen corpus already treats a food term used for something that is not the food. The remaining 126 rulings belong to the two passes described next.

The second pass exists because coder agreement is not self-certifying. Two coders who agree on a wrong include are never routed to adjudication, and the three boundary rules above were written down after the coding pass had run, so agreed includes could not have been judged against them. The author therefore re-read **all 380 agreed includes** against each record's own title and abstract rather than against the coders' stated reasons: **115 are excluded** under the vascular-bed, energy-expenditure and reactivity-probe rules, and 11 flagged by a keyword scan are confirmed as includes. A keyword scan alone was not sufficient and is reported as inadequate rather than used: it mis-sorted retinal microvascular density and optic-nerve-head microcirculation into the cutaneous tier on the word "microcirculation," and it flagged two records whose outcome is local thermal hyperaemia and which are therefore correct includes. Every override is filed in the published ledger with its rationale and appears in `screening.csv` as author-adjudicated, so no label changes silently. Before full screening, the author hand-labelled a 169-record golden set (29 include, 140 exclude) spanning the foods where the include boundary actually lives; against it, coder 1 achieved precision 0.867 / recall 0.897 / F1 0.881 and coder 2 precision 0.813 / recall 0.897 / F1 0.853. Gold labels for records whose abstract does not state the subject species were set from PubMed Humans/Animals Medical Subject Headings (MeSH) or full text, never by inference; this check corrected one gold label, and the correction was surfaced by a coder divergence rather than by the author's own review.

A third, adversarial pass over the foods that could flip. Two coders who exclude the same record for the same wrong reason leave no trace in κ, and because the primary outcome is binary, one such record would change a food's result outright. We therefore re-screened the records that stood to flip. The first pass read all **388 candidates belonging to the 23 core foods (≥ 3 Tier-1 sources) then at L2′ = 0** whose query returned anything at all; that scope was recomputed from the ledger rather than fixed, because widening the effect vocabulary changed which foods are core-zero and stranded an earlier pass built on the four-term set. A third agent (Claude Opus 5), blind to the existing labels, to the golden set and to the reconciliation code, was told the opposite of what the first two coders were told — that every record it was about to read had been excluded, and that its task was to find the ones where that was wrong. Its verdicts were `exclude-agreed`, `uncertain`, or `include-candidate`, with standing instructions to return `uncertain` whenever it hesitated and never to infer an unstated subject species. It returned **one include candidate and five uncertains**. The author read all six against title, abstract and MEDLINE indexing: the five uncertains stand as excludes, and **the include candidate holds** — a randomized double-blind crossover in twelve women whose posterior tibial artery blood flow rose significantly after ingesting wild watermelon juice, which the author's own earlier sweep had excluded by citing the reactivity-probe rule although the record contains no occlusion, no pharmacological probe and no flow-mediated dilatation. Before restoring it we checked the 39 limb- or skin-bed records that sweep excluded: every other one carries an organ or skeletal-muscle bed, or a named probe, so this record was the only one excluded on framing alone. The two remaining core-zero foods returned no L2 hits at all and so had nothing to re-screen; their zero is a question about the search vocabulary rather than about screening, which is where the Limitations place it.

Widening that pass to every food at zero. Scoping the check by coverage was itself a defect, and it is the one this revision repairs. Coverage is the analysis's explanatory variable, so re-reading only the high-coverage zeros makes the detection rate for a misclassified outcome a function of the variable under test; the record the first pass restored could only have been found where coverage was high, which is what makes the asymmetry material rather than theoretical. A second pass (protocol §9) therefore dropped the threshold and read **every remaining food in the primary frame at L2′ = 0 whose query returned at least one record: 1,331 candidates across 47 foods**, all of which carry one or two Tier-1 sources — the stratum where a correction moves the coverage estimate *down*, the direction the first scope could not produce. The instrument, the instruction and the verdict vocabulary were the first pass's, unchanged, and §2 and §3 gained no criterion; the 27 zero foods with no retrieved record remain outside both passes for the reason above. This pass flagged **three include candidates and four uncertains, and none was overturned.** Each of the seven already carried a standing reason in the ledger, written before the pass and withheld from it, and each resolves the same way on its own text: the bell pepper is the non-pungent control the chilli arms are measured against, in a trial whose conclusion attributes its thermogenesis to the chilli and the medium-chain triglyceride; the walnut study's question is substrate oxidation and its energy-expenditure result is null; the wakame trial is a weight-management study whose abstract says only "brown marine algae"; the white wine is held at an equal dose in both arms of a record that *is* included under `cheese`, because the cheese is what varies; the pear contrast is malabsorbers against absorbers; and two records are title-only. Every verdict and every ruling from both passes is published in `data/screening_thirdpass.csv`, each row tagged with the pass that produced it, and the instruction the screener received is published as `data/screening_thirdpass_coder_prompt.md`. That this second pass returned nothing is reported because §9 fixed in advance that it would be reported either way; a check run only until it stops finding things is not a check.

No publication-year restriction was applied at any layer; the retrieved records span the whole of PubMed's coverage, the oldest included here dating from the 1950s.

Auxiliary databases. OpenAlex (keyword-based) and the CiNii Research API (Japanese-language literature) were queried as auxiliary columns. OpenAlex `search=` is a full-text match and returns counts an order of magnitude larger than PubMed `[tiab]`, so the two are reported separately and never summed. The CiNii query retrieves the Japanese food name alone — an L1-equivalent total, **not** a thermal-context count; it therefore bounds how much Japanese-language literature exists on each food but does not establish that a food's thermal claim is unstudied in Japanese. The primary axis is the screened PubMed count.

Analysis

The analysis plan was fixed before any estimate was computed and is recorded in the project PLAN. The **primary model** is a logistic regression of whether a food has any on-construct study (L2′ > 0) on lay-source coverage (`n_sources`), adjusted for log(L1 + 1); the unadjusted fit is reported beside it. A binary outcome is what the data can support — L2′ is sparse, with most foods at zero and most non-zero foods at one or two studies — and "which foods have no direct research" is the paper's question. **Reported alongside** are Spearman rank correlations of coverage against L2′, against raw L2, and against L1, so that the effect of screening and the role of literature volume are both legible. The **sensitivity model** is a negative binomial regression of L2′ with log(L1 + 1) as an offset, fitted after a Poisson over-dispersion check (Green 2021; the model class has bibliometric precedent in Millones-Gómez et al. 2023). Branch conditions were fixed in advance: if the negative binomial failed to converge it would be reported as such rather than replaced by a zero-inflated model, which this many zeros over this few foods cannot identify; and a null primary result would be reported as no detected association rather than treated as a failed study.

Eighteen analyses are reported that were **not** in the fixed plan and are labelled as such wherever they appear. They are: the rank correlation of coverage against the L2′/L1 ratio; the refit of the primary model under the Tier 1+2 frame; four refits under narrower readings of what counts as an on-construct study — one dropping studies of a food's principal dietary constituent, one dropping reviews and meta-analyses, one dropping both, and one dropping the studies whose own question was not the food's thermal effect; three refits with coverage recounted within a single attribution vocabulary; two refits under narrower definitions of L1; three refits under other functional forms for the L1 adjustment; three complementary log-log refits treating literature volume as an exposure rather than a covariate — one imposing the proportionality that form assumes, one relaxing it, and one carrying the derivation's exact offset on the foods where that offset is defined; and a leave-one-food-out refit. All but the first two were added in revision. Each addresses a choice the primary specification had to make and could not settle from the data — what counts as an examination, what kind of attribution Axis A is counting, what L1 is a volume of, and what shape its adjustment assumes — so asserting any of them would leave the result resting on an untested judgment. All eighteen were computed after the primary estimates and none is used to support a conclusion.

Which attribution is being counted. Axis A assigns one warm/cool direction, but the sources reach it through three different vocabularies: a stated bodily effect ("体を温める", *warms the body*), a five-nature or yin–yang classification (温性/陽性), and a bare warm/cool food-category label (「温食材」, *warming ingredient*). Axis B measures a physiological outcome after ingestion, so counting a five-nature classification as an attribution that Axis B could examine is an identification this study assumes rather than demonstrates: that a food is classified 陰性 and that ingesting it lowers body temperature are not the same proposition. Each of the 394 Tier-1 quotations was therefore classified by its own wording and coverage recounted within each vocabulary. The classification is per quotation and not per source because the two vocabularies co-occur inside single lines — one source writes 「体を温める食材(陽性)＞根菜・冬野菜: にんじん」, carrying both at once — so 40 of the 74 five-nature quotations also state a bodily effect and no source-level partition separates them. The matching vocabularies were read off all 394 quotations rather than assumed; two quotations carry no thermal wording at all and were coded from surrounding text, and these are held in their own class rather than assigned to either. Whether a framing is fitted at all was fixed before any of these models ran, at ten events per predictor and two predictors: a framing with fewer than 20 foods having a study is described and left unfitted.

What L1 is a volume of. Because log(L1 + 1) is what carries the primary model, and L1 is a title-and-abstract count of the food name, the composition of that count varies by food — chicken's records are largely poultry science, salt's largely saline and sodium physiology. Two narrower definitions are therefore refit: L1 restricted to `humans[mh]`, and L1 restricted to the Medical Subject Headings nutrition tree (`"diet, food, and nutrition"[mh]`). Both filters were confirmed against the live index before use, PubMed returning them as MeSH terms with no unmatched-phrase warning, and both count snapshots are published. Separately, because 50 events over 146 foods is few enough that one food could carry an estimate, the primary model is refit 146 times with one food omitted each time, and the range of the coverage odds ratio across those refits is reported.

The zero-research rate is reported at every level of lay-source coverage rather than at selected thresholds, so that no cut-point is chosen after seeing the data. Warming and cooling foods are compared on the zero-research contrast (Fisher exact) and on L2′ (Mann–Whitney U) within the core set. All tests are two-sided at $\alpha = 0.05$; only the primary logistic model is confirmatory, and the remaining comparisons are descriptive and reported without correction for multiplicity. Ninety-five-percent confidence intervals use the Fisher z-transformation with the Bonett–Wright variance for rank correlations, **Wald bounds** for regression coefficients, Woolf's logit method for the odds ratio, and Wilson score intervals for the proportions in Figure 1. In the negative binomial the dispersion parameter is estimated by profiling the likelihood over a grid and then held fixed, so its interval is conditional on that value and is correspondingly narrow.

The two predictors are associated (Spearman $\rho = +0.480$), but not to a degree that destabilises the fit: on the model's own design matrix their Pearson correlation is +0.407 and each predictor carries a variance inflation factor of 1.20. The rank correlation is what Table 3 reports, but a variance inflation factor is defined from the linear fit between the predictors, so it is computed there rather than by squaring ρ. $\log(L1 + 1)$ rather than raw L1 enters the model, which compresses a range running from three foods holding no records at all to salt's 201,716 (hojicha has 29, milk 171,272) into a scale where no single food dominates. Beyond this we report no diagnostics of model fit: with 50 events over two predictors the model is used to describe an association, not to predict, and calibration or discrimination statistics would over-interpret it. The *functional form* of the adjustment is not treated as a diagnostic, because the reported result is a statement about what survives that adjustment. If the log-odds of having an on-construct study do not move linearly with $\log(L1 + 1)$, a single linear term leaves residual confounding, and the adjusted coverage estimate would then describe the shape of a mis-specified covariate rather than what the covariate does. The primary model is therefore refit three further times with coverage held as the term of interest and only L1's entry changed: as a quadratic in $\log(L1 + 1)$, as indicators for L1's tertiles — which assume no shape within a tertile and so cannot be led by one — and as a natural (restricted) cubic spline on $\log(L1 + 1)$ with three degrees of freedom. The number of parameters each specification spends is reported beside its estimate, because 50 events do not support many. A fourth variation changes the model rather than the term, and was proposed in review. If each of a food's L1 records were an independent opportunity for an on-construct study to exist, the probability that at least one does is $1 - \exp(-L1 \times \text{rate})$, which is a complementary log-log link carrying $\log L1$ as an offset — a form derived from what the outcome is rather than chosen for tractability. The offset carried is $\log(L1 + 1)$, so that all 146 foods enter one model with a finite exposure: three hold no records at all, and $\log L1$ does not exist for them. That substitution is an approximation to the exact form, so the exact form is fitted as well, on the 143 foods where $\log L1$ is defined. Dropping those three costs no information under the exact model, in which a food holding no records has probability zero of holding an on-construct study — which is what all three are. The approximation does not leave them silent: an exposure of one record makes their zeros bear on the estimate. The two offset fits therefore differ in what those three foods contribute as well as in the exposure they carry, and fitting both is what makes the size of that difference readable rather than asserted. It is fitted and reported three times: with the $\log(L1 + 1)$ offset, with that term free, and with the exact $\log L1$ offset. The offset *is* the assertion that volume's coefficient equals one, that assertion is testable, and the likelihood-ratio test between the 146-food offset and free fits is reported beside them, so that a reader can see whether the better-motivated form is one this data accepts before reading anything off it.

Analyses used Python 3.14.3 with pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, statsmodels 0.14.6, and matplotlib 3.10.8; the pipeline is covered by 451 unit tests (all passing). Every statistic in the Results and in Tables 3, 4 and 5, and the composition counts in Table 1, are recomputed from the pipeline output by `src/verify_stats.py`; Table 1's Axis A reliability figures come from `src/reconcile_coders.py`; the screening quantities in Table 2 — the κ, the agreement and adjudication counts, the golden-set scores and the exclusion-reason breakdown — are recomputed by `src/build_screening.py` from the published ledger, and the third-pass row by `src/build_thirdpass.py`. Figure labels are English-only to avoid Chinese–Japanese–Korean (CJK) glyph problems in the PDF build (weasyprint 68.1).

Results

Sample and the effect of screening

Axis A coding covered 301 distinct foods across 15 sources. Under the primary Tier-1 frame, 146 queryable foods carry at least one source attribution: 53 reach three or more sources (the *core* set), 25 have two, and 68 have one (Table 1). Among the 53 core foods, 33 are lay-classified warming, 19 cooling, and one contested.

Screening changed the measure substantially. Across the 146 foods in the primary frame, 11,714 raw L2 hits reduced to **284 on-construct studies** — a 97.6% reduction. Over the full 190-food query universe, on which the screening was carried out, 12,733 records reduced to 298 (Table 2), of which 271 are primary reports and 27 are reviews or meta-analyses of human thermal-ingestion evidence. Well over half of the 12,435 exclusions are animal research: 6,250 records excluded as `animal` and a further 858 as `livestock-heat`, together 57.2%. The next largest classes are `name-only` (2,441 records, 19.6%), covering studies where the food term matched but nothing about ingestion and body temperature was at issue, and `agri` (1,433 records, 11.5%) — crop cold tolerance and post-harvest physiology, which is the noise the wider vocabulary was accepted in order to buy recall.

The screening behaved as intended on the foods that motivated it. **Chicken fell from 702 raw hits to five**, a 99.3% reduction, with 566 of the 702 excluded as animal research. Of the five that survive, three concern chicken-derived preparations rather than the meat — an essence tablet, a hydrolysed sternal-cartilage extract, and a multi-ingredient supplement in which chicken essence is one component — leaving two trials of people eating chicken meat, both of nutritionally enriched meat assessed for microvascular reactivity. Lamb shows the pattern in its pure form (620 → 0). Ginger illustrates the opposite failure: 70 raw hits reduced to 12, and those 12 include the thermic-effect and energy-expenditure trials — Mansour et al. (2012), which the raw count had buried, and Fagundes et al. (2021), which the four-term query had not returned at all. Screening surfaces such studies; it cannot retrieve them, which is why the retrieval step had to be rebuilt rather than compensated for downstream.

The adversarial third pass over the 388 candidate records behind the current core-zero foods — 23 of the 25 then at zero had any record to re-read — returned one include candidate and five uncertains. The five uncertains did not survive the author's reading; the include candidate did (Methods), and **one food**

changed outcome: watermelon moves from no retrieved study to one, which is why 24 rather than 25 core foods appear in Table 4. A second adversarial pass then removed the asymmetry in that check itself. Extending it to the **1,331 excluded records of the 47 remaining zero foods** — the ones carrying one or two sources, which the first pass's coverage threshold had left out — flagged seven records and **overturned none**, so no food changed outcome and no estimate moved. The correction that had been available only where coverage was high was looked for where it was low, and was not there. Together with the author's sweep of every agreed include, which overturned 115 records in the other direction, these passes address the failure mode the agreement statistic cannot detect — two coders making the same mistake in the same direction — from both sides. None of them eliminates it, since every agent comes from one vendor's line.

No association was detected between lay-source coverage and whether a food has been studied

Of the 146 foods, **96 (65.8%) have no on-construct study**, and among the 53 core foods the rate is 24/53 (45.3%). Throughout what follows, "no on-construct study" is shorthand for none that this query retrieved and this screen retained. That is a floor on the truth rather than a census of it, for reasons the Limitations set out.

In the planned primary model, literature volume predicted whether a food had any on-construct study — **odds ratio (OR) 2.083 per unit of log(L1 + 1), 95% confidence interval (CI) 1.568–2.767, $p < 0.001$** — and lay-source coverage did not: **OR 1.062 per additional source, 95% CI 0.876–1.289, $p = 0.539$**; model pseudo-$R^2 = 0.288$ (Table 3). Unadjusted, coverage *is* associated with having been studied (OR 1.303, 95% CI 1.101–1.541, $p = 0.002$; pseudo-$R^2 = 0.054$). The association is therefore present in the data and is accounted for once literature volume is in the model, rather than being absent from the data to begin with — a distinction the earlier four-term version of this analysis could not draw, because there the unadjusted estimate did not reach significance either.

The coverage estimate is a non-finding rather than a demonstrated null. Compounded across the observed range of one to nine sources, its interval spans an odds ratio from 0.35 to 7.62 — that is, the data are compatible with widely covered foods being either substantially less or substantially more likely to have been studied. What can be said is that no association was detected, and that the same model with the same 50 events detected a clear one for literature volume.

The rank correlations show why adjustment matters. Coverage correlates positively with the screened count ($\rho = +0.331$, 95% CI +0.174 to +0.472, $p < 0.001$) — but it also correlates, more strongly, with the food's total literature ($\rho = +0.480$, 95% CI +0.336 to +0.602, $p < 0.001$). Widely covered foods are, on the whole, foods with large general literatures; once that is accounted for, the coverage association is not distinguishable from zero. Figure 1 shows this directly: the observed proportion studied rises with coverage, from 0.18 at one source to 0.64 at five or six (panel a), and rises monotonically with literature volume, from 0.08 to 0.69 across L1 tertiles (panel b), and when the primary model holds literature volume fixed the coverage curves are flat at every volume level while remaining widely separated from one another (panel c).

The correlation with raw, unscreened L2 was substantially higher ($\rho = +0.462$, $p < 0.001$) than with the screened count, indicating that about a third of the apparent coverage–attention association in the unscreened data was carried by records that do not measure the construct.

The zero-research rate falls as coverage widens and then stops falling. It is 65.8% across all 146 foods, 51.3% at two or more sources, 45.3% at three or more, and 44.7% at four or more; from there it flattens and rebounds on shrinking denominators — 48.0% at five or more, 52.6% at six or more, 63.6% at seven or more (Table 3). The confidence intervals in Figure 1a overlap from three sources upward, so the descent is carried by the move off the single-source level rather than by widening coverage within the core set. The per-food picture behind these rates is in Figure 2, where the 96 zero-study foods sit on the baseline across the whole coverage range rather than clustering at its narrow end.

The sensitivity model reached the same verdict of non-significance, and now a similar estimate. Poisson deviance/df was 4.715, confirming over-dispersion; the negative binomial with log(L1 + 1) as offset converged (dispersion parameter $\alpha = 3.010$) with a coverage incidence rate ratio (IRR) of 1.053 (95% CI 0.892–1.243, $p = 0.542$). In the four-term version of this analysis the offset model's coefficient was an order of magnitude larger than the primary model's and its interval nearly excluded unity; the two specifications now agree in estimate as well as in verdict. The offset still constrains the volume coefficient to exactly 1, a constraint the primary model's own estimate (0.734 on the log-odds scale) does not support — which is part of why the offset specification is reported as a sensitivity check rather than as the primary analysis.

Refitting the primary model under the wider Tier 1+2 frame (190 foods, an unplanned check) gave the same answer: 135/190 (71.1%) with no on-construct study, coverage OR 1.064 (95% CI 0.946–1.198, $p = 0.299$).

Narrowing what counts as an on-construct study did not change it either (all three checks unplanned). One hundred and three of the 284 studies examine a food's principal dietary constituent rather than the food itself; dropping them leaves 181 studies across 44 foods, a zero rate of 69.9%, coverage OR 1.050 (95% CI 0.867–1.270, $p = 0.620$) and volume OR 1.929 (1.474–2.524, $p < 0.001$). Dropping instead the 27 reviews and meta-analyses leaves 257 studies across 49 foods (66.4% zero), coverage OR 1.064 (0.878–1.291, $p = 0.525$). Dropping both — the two classes overlap in nine records — leaves 163 studies across 43 foods (70.5% zero), coverage OR 1.052 (0.869–1.273, $p = 0.606$) and volume OR 1.915 (1.464–2.504, $p < 0.001$). The narrowing is not spread evenly: it bites hardest on the foods carrying the most evidence, with green tea falling from 46 studies to 13 and coffee from 30 to 9, since both are substantially carried by constituent trials (Table 5). The estimates do not move.

A fourth narrowing puts the sharper question (also unplanned). Dropping the 120 of the 298 included records whose own framing was not the food's thermal effect — the `incidental` sub-label of protocol §8 — leaves 178, of which 171 fall in the primary frame, spread across 38 of its 146 foods, for a zero rate of 74.0%, coverage OR 1.086 (95% CI 0.890–1.324, $p = 0.417$) and volume OR 2.128 (1.564–2.897, $p < 0.001$). This is the narrowest reading of "examined" the data support and the one closest to *the claim was put to the test*; the coverage estimate moves by 0.024 and its interval still contains one. The sub-label reproduces, mechanically and over the whole ledger, the reading the chicken case forced by hand: of chicken's five surviving records the only claim-directed one is the essence-tablet trial of resting metabolic rate, and both enriched-meat trials are incidental. Salt shows the largest fall, from 19 studies to 4, most of its evidence being exercise-heat and vascular-reactivity work in which a thermal reading is a covariate; ginger is the opposite case, with all 12 of its studies claim-directed. Two limits belong with the figure. `claim-directed` records the study's *question*, not its motivation — almost nothing in this corpus cites a Japanese warming or cooling belief, and a criterion that required the citation would measure journal convention rather than what was studied — so even 171 overstates what has been addressed to the folk claim itself. And the sub-label is read from the title and stated aim, which makes it auditable but thin for multi-ingredient reviews, where whether a food is named in the framing or only in the body can turn on how the abstract happened to be written.

Nor did restricting coverage to one kind of attribution at a time (three unplanned refits). Of the 394 Tier-1 quotations, 305 state a bodily warming or cooling effect, 74 use five-nature or yin–yang vocabulary — 40 of them doing both in the same line — 53 are bare warm/cool category labels from a single source, and two carry no thermal wording and were coded from context. Counting coverage only from quotations that state a bodily effect leaves 125 foods, 44 with a study (64.8% zero), and gives coverage OR 1.154 (95% CI 0.901–1.478, $p = 0.257$) with volume OR 2.165 (1.562–3.000, $p < 0.001$); adding the category labels back, which is every attribution that is not five-nature theory, leaves 138 foods and gives coverage OR 1.095 (0.888–1.350, $p = 0.395$). Both keep the unadjusted association that the pooled model shows (OR 1.396, $p = 0.003$ and OR 1.320, $p = 0.003$), so the pattern of an association present before adjustment and absent after it survives the split. The five-nature frame is the one that cannot answer the question: only three Tier-1 sources use that vocabulary, so coverage there ranges over just one to two sources, and the refit — 59 foods, 21 with a study, coverage OR 0.782 (0.115–5.329, $p = 0.802$) — has an interval two orders of magnitude wide. It is reported because it was run, not because it discriminates; volume remains significant within it (OR 3.696, 1.774–7.699, $p < 0.001$). These three refits are counted on the frozen coding, which records each attribution at sentence granularity; the ledger's candidate spans are often a food name alone, with the wording that carries the vocabulary in the heading above it, so a vocabulary cannot be read from them. The two populations overlap without coinciding — 390 Tier-1 (source, food) pairs are common to both, one is in the frozen coding alone, and 107 are ledger additions that carry no framing label — so coverage counted within a vocabulary is a floor by that margin.

The volume adjustment does not depend on how L1 is defined, and no single food carries the coverage estimate (three further unplanned checks). Restricting L1 to human-indexed records cuts the frame's total from 1,448,752 to 346,923 — chicken from 94,310 to 19,480, salt from 201,716 to 51,058 — and restricting it to the MeSH nutrition tree cuts it to 422,070. Coverage stays null under both (OR 1.056, $p = 0.590$ and OR 1.045, $p = 0.665$), while volume becomes *stronger* rather than weaker (OR 2.575, 1.815–3.652 and OR 2.600, 1.822–3.709, both $p < 0.001$). Narrowing L1 towards the literature that studies these things as food therefore sharpens the term rather than dissolving it, which is the opposite of what a definitional artefact would do. Refitting the primary model 146 times with one food dropped each time, every refit converged and the coverage odds ratio stayed between 1.037 and 1.103, with p between

0.341 and 0.718 and no refit reaching significance. The four highest-volume foods moved it least of all — omitting salt, milk, egg or chicken shifted the odds ratio by at most 0.003 — and the largest shifts came from small foods with no study.

Nor does the estimate depend on the shape the adjustment assumes (three further unplanned refits). Against the primary model's 1.062, entering L1 as a quadratic in log(L1 + 1) leaves coverage at OR 1.072 (95% CI 0.878–1.309, $p = 0.493$); as indicators for its tertiles, at OR 1.126 (0.923–1.373, $p = 0.242$); and as a natural cubic spline with three degrees of freedom, at OR 1.074 (0.879–1.312, $p = 0.485$). The quadratic term is itself not significant ($p = 0.081$), so the evidence for curvature in the adjustment is weak rather than absent; what these refits establish is that allowing for curvature does not move the coverage estimate, which is the question a linear term raises. Every interval still contains one, and the most flexible of the three spends five parameters on 50 events — the limit of what this frame can be asked to carry.

Nor does it depend on treating volume as a covariate rather than as the exposure it arguably is (three further unplanned refits). Under the complementary log-log form, in which each of a food's L1 records is an opportunity, coverage carries a rate ratio of 1.036 (95% CI 0.892–1.202, $p = 0.647$) — the same verdict as the primary model, from a specification derived rather than chosen. The ratio is of latent rates per record, not of hazards over time, and not of risks: nothing here is measured on a clock. But that fit imposes the offset, and the offset asserts that volume's coefficient is exactly one. Refitting with the term free returns **0.606 (0.399–0.813)**, an interval that excludes one, and the likelihood-ratio test between the two specifications is **9.469 on 1 degree of freedom ($p = 0.002$)**. The proportionality the form assumes is therefore not one this frame supports: the latent rate per record falls as volume rises, so the latent expected number of on-construct studies rises sub-proportionally with volume rather than in step with it. Coverage sits in the same place under either version (with the term free, 1.054, 0.921–1.205, $p = 0.445$), which is why both are reported — the offset version alone would have rested a null result on an assumption the data decline to grant, and a null result is exactly where an unexamined assumption is easiest to accept. Carrying the derivation's exact offset instead, on the 143 foods for which log L1 exists, leaves coverage at 1.035 (0.892–1.202, $p = 0.648$); the approximation the reported fit makes is therefore not what produces the null. The same constraint is inside the negative binomial above, which is one reason that model is a sensitivity check rather than the primary analysis.

Widely covered foods with no retrieved study

Twenty-four of the 53 core foods have no on-construct study (Table 4). They include the foods most consistently presented as warming or cooling: carrot at all nine Tier-1 sources, cucumber and pumpkin at eight; burdock, eggplant, mango, and miso at seven. Their total literatures differ by more than two orders of magnitude — hojicha has 29 PubMed records and cucumber 12,174 — yet at neither extreme did the search retrieve a study of the thermal claim. Widening the effect vocabulary cut this set from 37 foods to 25, tomato and apple among those it moved out, and the adversarial third pass then removed watermelon, leaving 24.

Chicken is the sharpest case, and its shape changed when the query was widened. Six Tier-1 sources present it as warming; PubMed holds 94,310 records on it and 702 in the thermal keyword context; five of those 702 survive screening, and only two examine people eating chicken meat — both trials of nutritionally enriched meat assessed for microvascular reactivity, neither designed to ask whether chicken warms the body. Under the four-term query this count was zero and would have been reported as an absence. The wider vocabulary replaces the absence with a disproportion: roughly 94,000 records on the food, and two on people eating it in a thermal frame.

The asymmetry is in study design rather than in existence. Four of those 702 records put this very question to animals: the postprandial thermic effect of chicken in rats and its thyroid-hormone mediation (Wakamatsu et al. 2013), diet-induced thermogenesis across meat proteins in rats (Ezoe et al. 2016), diet-induced thermogenesis in broilers (Swennen et al. 2006), and — closest of all — a behavioural assay in which mice select their ambient temperature after eating different meats (Wakamatsu et al. 2025), which is the folk construct itself transposed to a rodent. All four are excluded here as animal research, correctly under the protocol, and each is identifiable by its exclusion reason in the published ledger. So the claim has been put to animals in designs built to ask it. In humans, the one retrieved study that set out to measure a thermogenic response to chicken used an essence tablet rather than the meat, and the two studies of people eating the meat measured microvascular reactivity, with local thermal hyperaemia among several vascular outcomes. That is a weaker statement than "the claim has not been put to humans," which is what the four-term query had licensed; it is also the statement the data support.

Warming versus cooling foods

Warming and cooling foods did not differ. Among core foods the zero-research rate was 14/33 (42.4%) for warming and 9/19 (47.4%) for cooling (Fisher exact OR 0.819, 95% CI 0.263–2.546, $p = 0.778$), and the screened counts did not differ (Mann–Whitney $U = 329.5$, $p = 0.757$; median 1 study in both groups). We make no claim of a warm/cool asymmetry.

Flagship foods

A handful of foods illustrate that "attention present" does not resolve the question (Table 5). **Green tea** (cooling, four sources) carries the most on-construct evidence in the whole frame with 46 studies, and **white sugar** (cooling, six sources) the second most with 37 — 0.30% and 0.04% respectively of their literatures — but 28 of green tea's 46 test a constituent and 17 are multi-ingredient formulations, so only 13 survive the narrowest reading. **Coffee** (cooling, four sources) has 30 on-construct studies from 23,286 records, yet 20 of them test caffeine or chlorogenic acid rather than coffee, and the best-powered relevant synthesis reports that caffeine *raises* core temperature under heat exposure (Peel et al. 2025), opposite to the lay attribution. **Ginger** (warming, eight sources) has 12, nine of them primary reports on ginger itself rather than on an isolated constituent, and those studies disagree: a thermic-effect difference at $p = 0.049$ in Mansour et al. (2012) alongside null total energy expenditure in the same trial, and null in Fagundes et al. (2021) — a trial that the four-term vocabulary had not retrieved at all. Even where the claimed context has been examined, it has been examined thinly, sometimes inconsistently, and in one case in the direction opposite to the belief.

Discussion

Across 146 foods that Japanese lay-facing sources classify as warming or cooling, the breadth of a food's coverage across those sources showed no detectable relationship with whether that food has been studied with an outcome that could speak to it. What that measures is how many independent lay-facing sources carry the attribution, not how widely it is held or how many readers meet it; the sources are a purposive sample of fifteen and are not weighted by readership, so "breadth" here is breadth of publication and the Limitations keep it there. The comparison that carries the paper is between foods holding similar amounts of general literature, and there an additional lay source corresponds to an estimated 6.2% change in the odds, with an interval spanning both directions. The positive association visible in the unadjusted data was substantially attenuated once literature volume was held fixed, which is what one expects if the same foods tend to be both widely written about and widely researched — though, as the Results note, the adjusted interval is wide enough that a real coverage effect of either sign remains possible.

Literature volume is in the model as the opportunity to be studied at all, and is not offered as a finding. L2′ is by construction a subset of the L1 records, so a food with a larger literature has more chances for at least one of its records to qualify, and the association between the two carries that part–whole coupling inside it. Reporting the volume term as a discovery would overstate what it is. What it earns is the comparison it makes possible: without it, coverage and volume are confounded at $\rho = +0.480$, and the unadjusted coverage association cannot be told apart from the fact that widely covered foods are, on the whole, widely studied foods.

This is a more specific reading than the one we set out to test, and it came from taking the measurement problem seriously. An earlier version of this analysis used the raw keyword co-occurrence count and reported a near-zero correlation between coverage and attention. That correlation was not interpretable, because well over half of the records screened — 7,108 of 12,733 — are studies of animals rather than of people eating food. After screening, the correlation is positive rather than null, and then is no longer distinguishable from zero once literature volume is held fixed, for a reason the raw analysis could not have identified. Construct validity was not a caveat on the result here; it determined what the result was.

The same applies one level down, to decisions the screening protocol had to make inside the construct. Whether a caffeine trial counts as evidence about coffee, and whether a review counts as an examination, are construct judgments rather than statistical ones. We tested both rather than asserting either, and neither moved the estimate. What the test did expose is that the narrowing falls unevenly: green tea carries the largest screened count in the frame and retains nothing under the strictest reading, because all of its evidence is about catechins or is second-hand. The aggregate result is robust to these choices; an individual food's count is not, and Table 5 should be read with that in view.

Three further judgments sit on the other axis and on the adjustment, and the same treatment applies to them. Axis A pools a stated bodily effect with a five-nature classification, which are not self-evidently one proposition; recounting coverage within each vocabulary separately left the estimate where it was, and left the unadjusted-to-adjusted attenuation intact, though the five-nature frame rests on three sources and is too imprecise to add much. L1, meanwhile, is what the result actually turns on, so its definition deserves the same scepticism as the outcome's: restricting it to human-indexed or nutrition-indexed records — which removes four-fifths of chicken's literature and three-quarters of salt's — strengthened the volume term instead of weakening it. That is worth stating plainly, because the alternative outcome was live: had the volume association depended on counting poultry science as research about chicken, the interpretation offered here would not survive. And since 50 events is few enough that one food could be doing the work, the model was refit without each food in turn; the coverage odds ratio never left the range 1.04 to 1.10, and the high-volume foods that motivated the check turned out to be the least influential of all.

The finding refines rather than contradicts the prior reviews. Ormsby (2021), Namiranian et al. (2021), and Zhou & Xu (2021) each establish that the field holds partial, mechanistically suggestive evidence, so "no evidence" is the wrong description; our contribution is orthogonal — we show that whatever evidence exists is associated with general research salience, with no association to how broadly a claim is listed detected once salience is held fixed, and that a standardized search and screen retrieved none at all for two-thirds of the foods these sources carry. Zhou & Xu's observation that food-level research is scarce relative to Chinese-medicine research is echoed quantitatively here: 12,733 records that mention a food alongside thermal vocabulary yield 298 studies of a human eating that food. García-Hernández et al. (2023) showed that a hot–cold belief system can be bibliometrically mapped and found domain-specific knowledge gaps in Mexico; our two-axis design extends that from documenting that a system exists to asking, food by food, what its claims' examination is associated with. Nagata et al. (2017) found that Japan's four warm/cool classification lists disagree when applied to cohort data; that internal disunity and the pattern reported here are complementary symptoms of a field elaborated culturally faster than it has been consolidated empirically.

One line of work deserves separate notice, because it is the closest thing to a sustained programme on this construct and it sits entirely inside our exclusions. Wakamatsu and colleagues have pursued the postprandial thermal effect of meat in rodents for over a decade: the thermic effect of chicken and its thyroid-hormone mediation (Wakamatsu et al. 2013), diet-induced thermogenesis across meat proteins (Ezoe et al. 2016), and most recently an assay in which mice choose their ambient temperature after eating different meats and their fractions (Wakamatsu et al. 2025) — a behavioural read-out of precisely the felt warmth the lay claim describes. The question this paper puts to the Japanese lay literature has therefore been asked, carefully and repeatedly, of rodents. What has not followed is the human study, and that is the shape of the gap these two axes describe.

The practical reading is diagnostic rather than causal. Research attention co-varies with a food's overall research salience, so the foods whose claims are least examined tend also to be those with the smallest general literatures: burdock, daikon, and hojicha are widely presented as warming or cooling and carry a few hundred PubMed records or fewer (323, 245 and 29 respectively). If that pattern holds, these claims are unlikely to meet evidence as a by-product of research on those foods for other reasons. Table 4 names 24 such foods with logged, reproducible counts, and they are the concrete targets a direct-verification programme would take up.

An important interpretive caution applies to caffeine. The evidence that caffeine raises core temperature comes from heat-exposure performance settings (sport, military, and industrial contexts) rather than from studies of everyday subjective warm/cool sensation, and physiological core-temperature change is a distinct measure from felt thermal comfort. We therefore treat the coffee reversal as a well-powered example of belief-incongruent physiological evidence in a specific context, not as a refutation of the subjective experience the belief describes.

Finally, the screening step is itself a transferable result — but a narrower one than a multi-model design would give. What we can claim is tractability: screening an entire 12,733-record hit set to a construct, twice over, with every judgment published, is now feasible at a scale that would previously have forced the sampling compromise this study was originally designed around. What we cannot claim is decorrelated error. Our two coders are adjacent capability tiers of one vendor's line, not the three-vendor arrangement of Hilkenmeier et al. (2026), so their 98.6% agreement ($\kappa = 0.8068$) bounds the internal consistency of that lineage rather than establishing independent convergence; comparing it to the human inter-rater $\kappa = 0.46$ reported by Guo et al. (2024) would also be misleading, since κ depends on prevalence and the two tasks differ sharply there — Guo et al. themselves report a prevalence- and bias-adjusted κ of 0.96 for their own comparison. The 176 divergences were where the difficulty concentrated, and one of them exposed an error in the author's own reference labels: disagreement between models was useful as a flag for hard records even though agreement between them cannot be read as independent confirmation.

The sharper methodological lesson is that agreement is not self-certifying in either direction, and that both directions had to be checked because both turned something up. Of the 380 records both coders included, 115 were wrong under boundary rules written down after the coding pass had run — chiefly endothelial-function trials whose vascular outcome is not a thermal one. No agreement statistic could have surfaced them, because the two coders agreed; only an author pass over every agreed include did. Running the mirror check on the excludes then produced the sharper result, because what it overturned was not a coder's label but the author's own: a record the include sweep had excluded by citing a rule whose named triggers it does not contain. An adversarial pass is usually justified as a check on the coders. Here it worked as a check on adjudication, which is the step no agreement statistic covers at all and the step a reader has least ability to audit. Any pipeline that routes only divergences to human adjudication inherits the first blind spot; any pipeline that treats author rulings as final inherits the second.

Limitations

First — lay-source coverage is a source count, not a measure of conviction. Axis A counts how many independent sources publish an attribution; it does not weight by readership, and appearing in many sources is not the same as being widely believed. The frame is 15 Japanese lay-facing web sources frozen in advance and dominated by corporate and association wellness media. A different frame — print encyclopedias of *yakuzen*, or reader surveys — could shift individual foods' values. Widening the frame to include the six individual-expert blogs did not change the conclusion (an unplanned check; OR 1.064, $p = 0.299$). The measure also pools attributions that are not obviously one construct: most quotations state a bodily effect, but 74 classify the food under the five natures or yin–yang, and that a food is 陰性 is not the proposition that eating it lowers body temperature. Recounting coverage within each vocabulary left the result unchanged where it could be tested, but the five-nature frame draws on only three sources and its interval is too wide to discriminate, so the question is bounded rather than settled: what we can say is that the finding is not an artefact of pooling, not that the two vocabularies have been shown to name the same thing.

Second — the coders were language models, and neither reliability statistic supports an independence claim. All coding on both axes was performed by large language models under the author's direction; there were no human raters. Both axes pair two capability tiers of one vendor's line — Sonnet with Opus — so Axis A's $\kappa = 0.8998$ over 18,730 candidate decisions and Axis B's $\kappa = 0.8068$ over 12,733 records are within-lineage consistency estimates rather than evidence of independent convergence; models sharing pretraining data and alignment methodology should be expected to share failure modes. Neither figure is comparable to a κ between trained human screeners, and κ is in any case prevalence-dependent, which Axis B's 2.3% include rate makes acute. The earlier hand coding of Axis A reported $\kappa = 1.000$, and that figure covered one thing only — the direction assigned to items both coders had already extracted — while extraction, meaning which foods to record at all, was where those coders actually diverged, at a Jaccard index of 0.54 in its second round — a round that, unlike the rest of the study, ran both coders as separate agents of one model rather than as two tiers. That pass remains live as the reference set against which the enumeration's recall and the attribution-vocabulary split are measured, so its weaker pairing is stated here and not only in the protocol. Giving both coders one enumerated candidate set removes that gap, so the κ reported here is computed over the decision that matters and over every candidate; it is lower than the earlier figure because it measures a harder and more complete thing. What does not depend on model agreement is the verbatim grounding check on every Axis A row, the golden-set precision and recall against hand labels, and the publication of every judgment on both axes with its reason, all of which can be audited directly. The failure mode that no agreement statistic can detect — both coders reaching the same wrong label — was addressed from both sides. On the include side, the author's sweep over all 380 agreed includes overturned 115. On the exclude side, a third pass instructed to overturn rather than to concur read all 388 candidate records of the then core-zero foods and overturned one — and that one had been excluded not by the coders but by the author's own sweep, on a rule whose named triggers the record does not contain. Because that pass was scoped by coverage, which is the analysis's own explanatory variable, it was extended to every remaining food at zero — 1,331 records across 47 foods, all carrying one or two sources — where it overturned nothing. So the failure mode is demonstrated rather than hypothetical in both directions, and the exclude-side check additionally caught an adjudication error, which no agreement statistic covers. Both bound the concern without eliminating it, since every agent here is drawn from the same vendor's line and the author is a single reader. On Axis A the concern was also measured directly, under the procedure fixed in advance and described in the Methods. Of **60 claim rows drawn at random**, **59 are supported** by their source and one is not: a row recorded as 野菜 (*vegetables*) where the source says 冬が旬の野菜 (*vegetables in season in winter*) — the attribution is real and the label is broader than it. Two sources were then read end to end without reference to the ledger. Counted as specified, where every name the ledger does not hold under the reader's own surface form is missed, recall is **43/71** for the longer source and **37/40** for the other. Adjudicated against the coding rules, most of those names are ones the ledger holds under another form (肉類 for 肉, 根菜類 for 根菜) or ones it was right to omit — serving temperature, generic terms, and dishes of which no direction is predicated — leaving **53/56** and **37/37**, and **three genuine gaps**, all in the longer source: おでん (*oden*), named beside 豚汁 (*pork miso soup*) which the ledger does hold; 発酵茶 (*fermented teas*), a class of which it holds two members; and 精製された白い食品 (*refined white foods*), a class of which it holds one. Both figures are reported because giving only the second would be choosing a number after seeing it. All four defects are class labels or prepared dishes, the categories the query stage already excludes — 72 of the 218 Tier-1 foods — so on that same rule none of them enters the 146-food analysis frame, and none moves `n_sources` for any food analysed here. Sixty rows and two sources of nine can detect a systematic failure; they cannot certify the absence of one.

Third — the screened count is a floor, not a census. L2′ counts studies that the 22-term PubMed query retrieved *and* that survived screening. Screening is a filter, so it can only remove; a study phrased entirely in other vocabulary never enters the candidate set and no amount of screening recovers it. That this failure is real rather than hypothetical is demonstrable from this study's own history. The first version of this analysis used four effect terms, which was narrower than the inclusion rule the screening applied, and it missed Fagundes et al. (2021) — a randomized crossover trial of ginger measuring thermic effect of food, indirect calorimetry and axillary temperature — entirely. Widening the vocabulary to match the inclusion rule retrieved it, and brought in enough further evidence to cut the core-zero set from 37 foods to 25 and the overall zero rate from 80.8% to 65.8%. This also puts a number on the earlier query's recall. Taking as the reference set the 292 studies the 22-term query retrieved and screening retained when that comparison was run — the ledger now holds 298, the six added by a later round of screening rather than by any change to the query — the four-term query had returned **104 of them, a recall of 35.6%**, so 188 on-construct studies were outside its reach, and any absence it reported was an absence at that recall. The record the third pass restored is one of the 188: the four-term query never returned it. The correction is instructive precisely because it can recur: the 22 terms are matched to our own inclusion rule, not to the full space of ways a thermal effect can be reported, so L2′ still under-counts by an unknown margin. Zero means "no study was retrieved by this query and retained by this screen," not "no such study exists."

Fourth — database coverage is incomplete and the Japanese-language check is weaker than it should be. PubMed does not index J-STAGE/CiNii, KMbase/OASIS, or CNKI, so the counts under-represent Japanese, Korean traditional-medicine, and Chinese literatures. Our CiNii query returns total counts for the Japanese food name, not thermal-context counts, so it bounds the size of the Japanese literature on each food but cannot confirm that a food's thermal claim is unstudied in Japanese. A thermal-context search of Japanese-language databases is the most direct extension of this work.

Fifth — search terms are not equally apt for every food. Some synonyms conflate distinctions the belief makes (green tea and black tea share *Camellia sinensis*), and the unit of the query does not always match the unit of the claim (lay sources speak of chicken breast; the query returns work on the whole species). Screening removes the resulting false positives but cannot recover a study the query never retrieved, so residual differences in query aptness remain a source of between-food measurement error.

Sixth — cross-sectional snapshot. Both axes were measured once, in August 2026. Lay sources are edited and PubMed counts grow, so specific values will drift, although the structural pattern is unlikely to reverse on the timescale of source updates.

Seventh — descriptive, ecological, unregistered, and imprecise. The design identifies an association pattern across foods, not a mechanism or a cause, and food-level associations should not be read as statements about individuals. The analysis plan was fixed before estimation but was not registered on a public protocol registry, and eighteen reported analyses were computed outside it and are labelled where they appear. Precision is the binding constraint on the central result: 50 of 146 foods have any on-construct study, and the coverage interval compounds across the observed one-to-nine-source range to an odds ratio between 0.35 and 7.62. The absence of a detected coverage effect is therefore a non-finding, not evidence that no effect exists, and a frame with more sources or a database with more on-construct studies could resolve it either way.

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Ethical considerations

This study used exclusively publicly available information: aggregate bibliographic counts and public bibliographic records from public literature databases (PubMed, CiNii, OpenAlex) and publicly published lay-media web pages. No individual-level, health, or personally identifying data were accessed, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public bibliographic data and public web content, it does not meet the standard threshold for human-subjects research under either domestic (Japanese Ethical Guidelines for Medical and Biological Research Involving Human Subjects, 2021 revision, for research using publicly available information) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required. The study was conducted in accordance with the principles of the Declaration of Helsinki where applicable to research using aggregate, publicly available data.

Acknowledgments

Large language models were used as instruments in this study, not only as writing aids, and their roles are separated here accordingly.

As coding instruments. Axis A was coded twice in the study's history, and both passes are published. The frozen reference set was coded by Claude Opus 4.8 with Claude Sonnet 4.6 (Anthropic), except for its second round, which ran both coders as separate agents of Claude Sonnet 4.6 rather than as two tiers. The candidate ledger analysed here — the pass whose κ = 0.8998 over 18,730 candidates is the figure reported

throughout this paper — was coded by Claude Sonnet 5 with Claude Opus 5. Axis B abstract screening was performed twice independently by paired models of adjacent capability tiers — Claude Sonnet 4.6 with Claude Opus 4.7, and Claude Sonnet 5 with Claude Opus 5 — under the protocol in `data/screening_protocol.md`. Both adversarial third passes over the foods at zero were performed by Claude Opus 5. All models are from a single vendor; the Methods and Limitations state what that does and does not license. Every divergence, every record flagged for missing information, and every coverage difference was adjudicated by the author against the primary record. Model versions differ across batches because the work spanned several sessions; the batch-level assignment is tabulated in the protocol files.

As development and writing aids. Claude Opus 4.8 and Claude Opus 5 (Anthropic) assisted with code drafting, debugging, literature-search support, and manuscript editing.

All data extraction, statistical outputs, and numerical results were independently checked by the author by re-executing the analysis code (Python 3.14.3; unit tests n = 451, all passing) and by recomputing every reported statistic directly from the pipeline output before writing. The author is solely responsible for the accuracy of the numerical results, the choice of specifications, the interpretation, and the conclusions. All references were verified against PubMed, CrossRef, and publisher records; DOIs and PMIDs were confirmed against primary source pages, and two cited works (Jin et al. 2025 and Miyoshi et al. 2026) are not indexed in PubMed and are cited by DOI.

Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Axis B counts derive from the NCBI PubMed E-utilities (https://www.ncbi.nlm.nih.gov/books/NBK25501/), the CiNii Research API (https://support.nii.ac.jp/en/cinii/api/api_outline), and OpenAlex (https://openalex.org/). Query strings are generated deterministically by the published query builder (`src/food_query_terms.py`, `src/`

evidence_mapping.py) from each food's key, so every query in the study can be regenerated exactly; the 1,345 raw count responses that those queries returned are published under data/query_log/, pinning each reported count to its retrieval snapshot. The two narrower L1 definitions used in the covariate sensitivity check are built by src/alt_l1.py and their per-food counts published as data/alt_l1_counts.csv, each row carrying the query string that produced it. The one class of raw response **not** redistributed is the PubMed efetch dumps of titles and abstracts, which are third-party content; they are regenerable with src/fetch_l2_records.py, and every PMID they contain appears in the published screening ledger.

Axis A source pages are the 15 publicly published lay-media URLs listed in data/sources.csv with their access dates; the raw HTML is not redistributed, and each coded row in data/claims.csv carries the verbatim supporting quotation, so every attribution can be located in the live source. The full screening ledger — one row per (food, PMID) with both coders' labels, whether the author adjudicated it, the final label, and the reason code — is published as data/screening.csv, the hand-labelled reference set as data/screening_golden.csv, and the two coding protocols as data/coding_protocol.md and data/screening_protocol.md. The §8 sub-label pass is published in the same form: data/screening_claim_directed.csv carries one row per included record with both coders' sub-labels, whether the author adjudicated it, the settled sub-label and the coder's statement of the study's own question; data/screening_claim_directed_rulings.csv carries the author's 34 rulings with their rationales; and data/screening_claim_directed_coder_prompt.md carries the instruction the coders received. The adversarial third passes are published as data/screening_thirdpass.csv, one row per re-screened record carrying its verdict, its reason, the pass that produced it, and the author's ruling where one was required; the instruction the screener received is published as data/screening_thirdpass_coder_prompt.md. The Axis A candidate ledger — one row per (source, span) with both coders' labels, whether the author adjudicated it, the final label, the reason code and, for included rows, the food name and direction — is published as data/claims_ledger.csv, the author's rulings with their rationales as data/ledger_rulings.csv, and the prompt the coders received as data/ledger_coder_prompt.md. The earlier hand coding is retained unchanged as data/claims_frozen.csv, because the enumeration's recall and the attribution-vocabulary split are both measured against it. The human audit of §10 is published in full: the drawn sample as data/human_audit_sample.csv, the author's verdict and note per row as data/human_audit_results.csv, the two end-to-end reads as data/human_audit_source_reads.csv, and the adjudication of every name those reads raised, with its reason, as data/human_audit_rulings.csv. The fifteen independent full-source reads used to check the enumeration without reference to any coding are published as data/independent_read/*.json.

Analysis code (Python 3.14.3, pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, statsmodels 0.14.6, matplotlib 3.10.8, weasyprint 68.1) — including the source-fetching and text-extraction scripts, the claim-coding and Axis A aggregation modules, the Axis B query builders and PubMed/CiNii/OpenAlex collectors, the record-fetching and screening modules, the query logs, the framing and alternative-covariate modules, the figure pipeline, and the verification scripts that between them recompute every number reported here (`src/verify_stats.py` for the analysis statistics, `src/build_ledger.py` for the Axis A ledger and its agreement figures, `src/verify_candidate_recall.py` and `src/verify_independent_read.py` for the completeness of the enumeration, `src/build_screening.py` for the screening reliability and exclusion counts, `src/build_claim_directed.py` for the §8 sub-label ledger and its agreement figures, `src/build_thirdpass.py` for the third-pass ledger, and `src/verify_manuscript.py`, which walks every number printed here and checks it against those outputs) — is publicly available at https://github.com/rehabilitation-collaboration/warming-cooling-foods (MIT License). All numerical results reported in this manuscript are reproducible from the analysis code and published ledgers at repository commit `ac02384`, recorded as the reproducibility anchor for this revision, with one class of exception: the enumeration-completeness figures in the Axis A methods — the candidate and occurrence counts, the body-text and thermal-line coverage, and the recall measured against the frozen coding — are computed from the fetched source pages, which are not redistributed, so `src/verify_candidate_recall.py` and `src/verify_independent_read.py` cannot be re-run from the public repository alone (the ledger's own 18,730 candidate total is recoverable from `claims_ledger.csv`); the anchor for the first posting was `83394075d4ab93e60393a9aa8595a78dfe0b2a77`, which reproduces the earlier four-term measure rather than the one reported here.

Tables

Table 1. Sample composition

Set	Definition	n foods	Warming	Cooling	Contested/neutral
All coded	≥ 1 source, any tier	301	—	—	—
Tier-1 covered	≥ 1 Tier-1 source	218	—	—	—
Analysed (primary frame)	**queryable, ≥ 1 Tier-1 source**	**146**	**78**	**60**	**8**
— core	≥ 3 Tier-1 sources	53	33	19	1
— two-source	2 Tier-1 sources	25	—	—	—
— single-source	1 Tier-1 source	68	—	—	—

Coding covered 301 distinct foods across 15 sources (Tier 1: 9 corporate/association; Tier 2: 6 individual-expert blogs) in 824 rows, generated from a ledger of 18,730 candidates judged independently by two coders at Cohen's κ = 0.8998 with 208 author adjudications (Methods). Of the 218 foods with Tier-1 coverage, 72 are labels no single unambiguous query can represent — 54 classes ("leafy greens," "spices," "nuts," "shellfish" and the like), 16 prepared dishes, and two composite descriptions ("fatty meat," "mixed nuts") — and were excluded from Axis B. The 44 queried foods with no Tier-1 source lie outside the primary frame and enter only the Tier 1+2 sensitivity refit.

Table 2. Effect of construct screening on the research measure

Quantity	Value
L2 records screened (all queried foods)	12,733
Foods with ≥ 1 L2 hit / queried with none	146 / 44
Two-coder Cohen's κ (co-coded n = 12,733)	0.8068
Observed agreement	98.6%
Coder divergences	176
Records adjudicated by the author	499
— divergences and information-gap flags / §2.3 sweep of agreed includes	217 / 125
— reason sweep of agreed exclusions (one food's homonyms)	156
— third-pass restoration of one swept include	1
Records surviving as on-construct (L2′)	**298**
— primary reports / reviews	271 / 27
Raw L2 → L2′ within the primary frame	11,714 → 284 (−97.6%)
Author sweep of agreed includes: records / excluded / confirmed	380 / 115 / 11
Third-pass re-screen, core-zero foods: records / flagged / overturned	388 / 6 / 1
Third-pass re-screen, remaining zero foods: records / flagged / overturned	1,331 / 7 / 0

Exclusion reasons (n = 12,435):

Reason	n	Share
`animal` (non-human subjects)	6,250	50.3%
`name-only` (food named, no thermal-ingestion question)	2,441	19.6%
`agri` (crop cold tolerance, post-harvest physiology)	1,433	11.5%
`livestock-heat` (farm-animal heat stress)	858	6.9%
`no-thermal` (human ingestion, no thermal outcome)	761	6.1%
`invitro`	319	2.6%

Reason	n	Share
`not-ingestion` (non-oral administration)	180	1.4%
`species-mismatch` (different species in the same family)	167	1.3%
other (`mechanism-only`, `uncertain-species`)	26	0.2%

L2 is the raw count of PubMed records in which a food term co-occurs with one of the 22 thermal-effect terms; L2′ is the subset that screening retained as on-construct. The §2.3 sweep disposed of 126 records, but the 115th exclusion is the one the third pass restored and is counted in that row instead, so 125 rulings are filed under the sweep itself. The reason sweep covers 156 `cream` *records whose label was left untouched and whose reason code alone was corrected. Golden-set validation (169 hand-labelled records: 29 include, 140 exclude): coder 1 precision 0.867 / recall 0.897 / F1 0.881; coder 2 precision 0.813 / recall 0.897 / F1 0.853.*

Table 3. Primary and supporting statistics (primary frame, n = 146)

Analysis	Estimate	95% CI	p
Logistic, adjusted — lay-source coverage	**OR 1.062**	**0.876 to 1.289**	**0.539**
Logistic, adjusted — log(L1 + 1)	**OR 2.083**	**1.568 to 2.767**	**< 0.001**
Logistic, unadjusted — coverage	OR 1.303	1.101 to 1.541	0.002
Spearman ρ — coverage vs L2′	+0.331	+0.174 to +0.472	< 0.001
Spearman ρ — coverage vs raw L2	+0.462	+0.316 to +0.587	< 0.001
Spearman ρ — coverage vs L1	+0.480	+0.336 to +0.602	< 0.001
Spearman ρ — coverage vs L2′/L1	+0.297	+0.138 to +0.442	< 0.001
Negative binomial (offset log(L1+1)) — coverage	IRR 1.053	0.892 to 1.243	0.542
Logistic, adjusted — coverage, food-form studies only	OR 1.050	0.867 to 1.270	0.620
Logistic, adjusted — coverage, primary reports only	OR 1.064	0.878 to 1.291	0.525
Logistic, adjusted — coverage, both narrowings applied	OR 1.052	0.869 to 1.273	0.606
Logistic, adjusted — coverage, L1 as a quadratic	OR 1.072	0.878 to 1.309	0.493
Logistic, adjusted — coverage, L1 as tertile indicators	OR 1.126	0.923 to 1.373	0.242
Logistic, adjusted — coverage, L1 as a cubic spline	OR 1.074	0.879 to 1.312	0.485
Fisher exact — warming vs cooling zero rate (core)	OR 0.819	0.263 to 2.546	0.778
Mann–Whitney U — warming vs cooling L2′ (core)	U = 329.5	—	0.757

Zero-research rate at each level of lay-source coverage:

Coverage ≥	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Foods	146	78	53	38	25	19	11	6	2
No on-construct study	96	40	24	17	12	10	7	3	1
%	65.8	51.3	45.3	44.7	48.0	52.6	63.6	50.0	50.0

Primary model pseudo-R^2 = 0.288; 50 of 146 foods have ≥ 1 on-construct study. The unadjusted model's pseudo-R^2 is 0.054. Regression intervals are Wald. Poisson deviance/df = 4.715 confirmed over-dispersion before the negative binomial was fitted; its dispersion parameter (α = 3.010) was estimated by grid-profiling the likelihood and then held fixed, so the IRR interval is conditional on that value. The two foods at coverage ≥ 9 — banana with two on-construct studies and carrot with none — make that column uninformative on its own; it is shown because the analysis plan undertakes to report every level rather than selected thresholds. Eighteen analyses were computed outside the fixed plan and are labelled as such wherever they appear. The three functional-form refits change only how L1 enters — a quadratic in log(L1 + 1), tertile indicators, or a natural cubic spline with three degrees of freedom — and spend 4, 4 and 5 parameters respectively against the primary model's 3; the quadratic term is not significant (p = 0.081). The three complementary log-log refits instead treat L1 as an exposure, and their ratios are of latent rates per record rather than of hazards: with log(L1 + 1) as an offset, coverage 1.036 (0.892–1.202), p = 0.647; with that term free, 1.054 (0.921–1.205), p = 0.445, its coefficient 0.606 (0.399–0.813), and the likelihood-ratio test of the offset against it 9.469 on 1 df, p = 0.002; with the derivation's exact log L1 offset, fitted on the 143 foods for which it is defined, coverage 1.035 (0.892–1.202), p = 0.648. The proportionality test is reported for the 146-food log(L1 + 1) offset/free pair; the exact log L1 fit is reported only as a sensitivity check on the +1 approximation. The Tier 1+2 refit (n = 190) gave coverage OR 1.064 (0.946–1.198), p = 0.299, with 135/190 (71.1%) having no on-construct study; and the three narrowed-definition refits are the rows above, which drop the 103 studies of a principal dietary constituent, the 27 reviews and meta-analyses, or both (the classes overlap in nine records), leaving 44, 49 and 43 foods respectively with any study (zero rates 69.9%, 66.4% and 70.5%). Volume stayed significant under every narrowing (OR 1.929, 2.061 and 1.915; all $p < 0.001$). Recounting coverage within one attribution vocabulary at a time gave coverage OR 1.154 (0.901–1.478), p = 0.257 on the 125 foods attributed a bodily effect, OR 1.095 (0.888–1.350), p = 0.395 on the 138 attributed anything other than a five-nature classification, and OR 0.782 (0.115–5.329), p = 0.802 on the 59 carrying a five-nature classification, where only three sources use that vocabulary. Refitting under L1 restricted to human-indexed records gave coverage OR 1.056, p = 0.590 and volume OR 2.575 (1.815–3.652); under L1 restricted to the MeSH nutrition tree, coverage OR 1.045, p = 0.665 and volume OR 2.600 (1.822–3.709). Across 146 leave-one-food-out refits the coverage odds ratio ranged from 1.037 to 1.103 and no refit reached $p < 0.05$. Only the adjusted logistic model is confirmatory; the rest are descriptive and uncorrected for multiplicity.

Table 4. The 24 core foods for which the search retrieved no on-construct study, by lay-source coverage

Food	Lay direction	Sources	Total literature (L1)	Raw L2	On-construct (L2′)
carrot	warming	9	5,702	12	0
cucumber	cooling	8	12,174	67	0
pumpkin	warming	8	2,631	15	0
mango	cooling	7	3,627	14	0
eggplant	cooling	7	2,072	13	0
miso	warming	7	1,158	15	0
burdock	warming	7	323	0	0
melon	cooling	6	3,898	35	0
pineapple	cooling	6	2,494	10	0
green onion	warming	6	401	1	0
onion	warming	5	7,727	30	0
tofu	cooling	5	1,188	3	0
tuna	warming	4	3,715	76	0
brown rice	warming	4	2,040	5	0
white rice	cooling	4	1,514	3	0
kimchi	warming	4	1,180	6	0
daikon	contested	4	245	0	0
sweet potato	warming	3	3,941	15	0
buckwheat	warming	3	2,799	7	0
chinese cabbage	cooling	3	2,115	11	0
prune	warming	3	1,957	10	0
soy sauce	warming	3	1,264	2	0
kiwi	cooling	3	1,195	6	0
hojicha	warming	3	29	1	0

"Raw L2" is the unscreened keyword co-occurrence count. Every one of these 24 foods has at least three independent Tier-1 sources assigning it a thermal direction, and none of the records this query retrieved and this screen retained studies a human ingesting it with a thermal outcome measured. Widening the effect vocabulary from four terms to 22 cut this set from 37 foods to 25, and the adversarial third pass

then removed watermelon, so a zero here is a floor on the evidence rather than a census of it: a study phrased entirely outside even the 22-term vocabulary never entered the candidate set (Limitations, third point).

Table 5. Flagship foods: total volume, keyword hits, and on-construct studies

Food	Lay direction	Sources	L1 total	Raw L2	L2′	Food-form primary	L2′/L1	What the direct evidence shows
chicken	warming	6	94,310	702	5	3	0.005%	566 of the 702 hits are avian or poultry-production research; of the five that survive, three test chicken-derived preparations and two test people eating enriched chicken meat, both measuring microvascular reactivity
green tea	cooling	4	15,476	256	46	13	0.297%	The largest screened count in the table, but 28 of the 46 test a constituent and 17 are multi-ingredient formulations flagged `confounded`; 13 survive the narrowest reading
white sugar	cooling	6	85,454	812	37	15	0.043%	Second-largest count and the more widely covered food; a mixture of thermic-effect-of-food trials and taste-stimulus studies recording skin blood flow and temperature
coffee	cooling	4	23,286	165	30	9	0.129%	Caffeine *raises* core temperature under heat exposure (Peel et al. 2025, n = 30, g = 0.44) — opposite to the lay attribution — and 20 of the 30 test caffeine or chlorogenic acid rather than coffee
ginger	warming	8	5,730	70	12	9	0.209%	The most-covered food in the frame and the one with the most evidence on the food itself rather than on an isolated constituent: a thermic-effect difference at p = 0.049 in Mansour et al. (2012, n = 10) — with total resting energy expenditure and the thermic-effect AUC both null at p = 0.43 in the same trial — null in Fagundes et al. (2021, n = 20), and null again for thermoregulatory function on repeated dosing

Food	Lay direction	Sources	L1 total	Raw L2	L2′	Food-form primary	L2′/L1	What the direct evidence shows
chili pepper	warming	6	1,943	50	17	9	0.875%	Seven of the 17 are capsaicin or capsinoids at dietary doses; the highest on-construct share among the foods in this table, though not in the frame (mugicha reaches 8.11% on three studies out of 37 records)
milk	cooling	3	171,272	1,932	17	11	0.010%	1,479 of the 1,932 keyword hits are animal or livestock-heat research; the largest total literature in the study and one of its smallest on-construct shares

"Food-form primary" is an unplanned quantity, added in revision: the count remaining when records sub-labelled as testing a principal dietary constituent and records sub-labelled as reviews or meta-analyses are both removed — the narrowest reading reported in the Results. A preparation made from the food — an extract, powder or tablet of it — is a food-form under the screening protocol rather than an isolated constituent, and so stays in this column; chicken's 3 therefore includes an essence tablet and exceeds the two trials of people eating the meat. The two classes overlap, so the column is not L2' minus their sum. L2'/L1 is computed on L2' as reported. Raw L2 and L2' are both on the 22-term query; under the earlier four-term query chicken returned 278 hits and no surviving study.

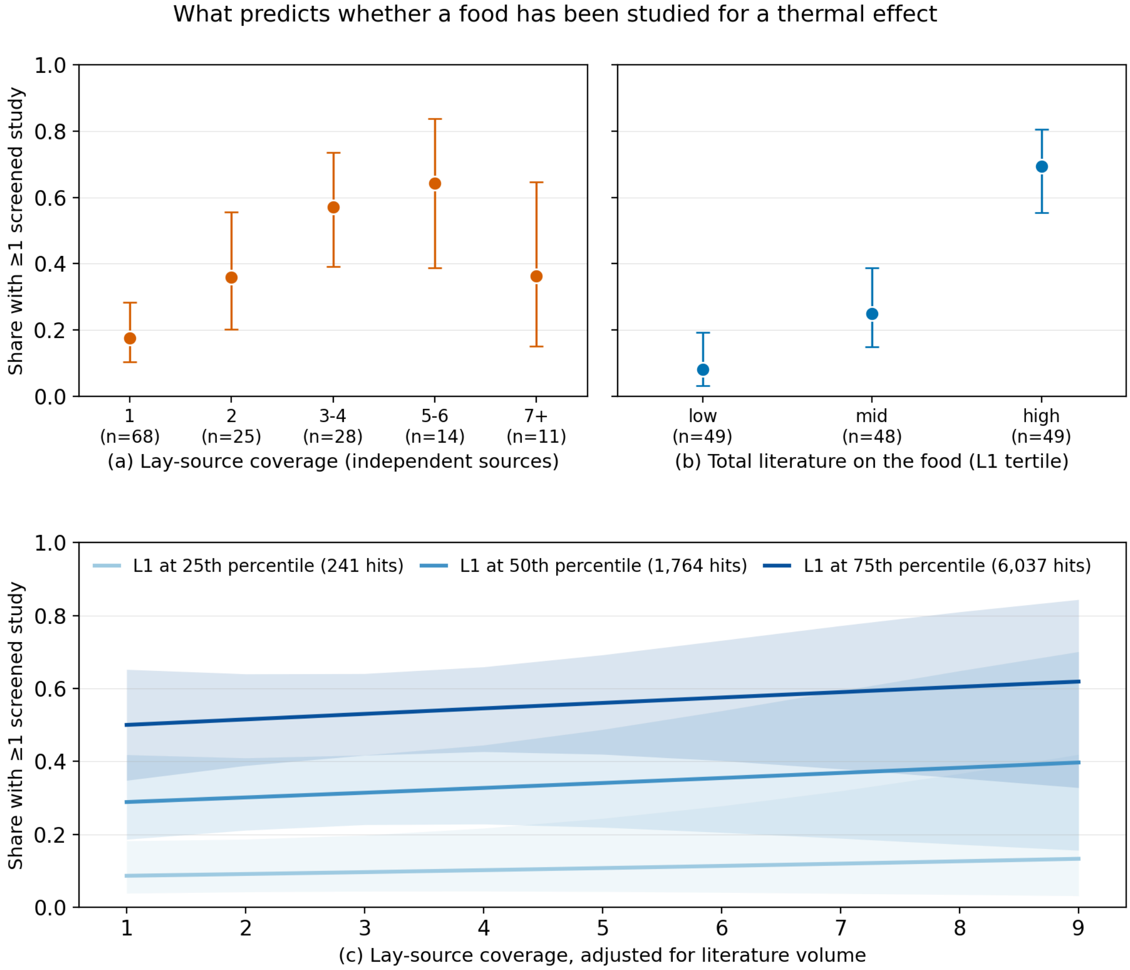

Figure 1. What predicts whether a food has been studied for a thermal effect (n = 146). The outcome throughout is the share of foods with at least one on-construct study (L2′ ≥ 1), on a common y axis. **(a)** Observed share by lay-source coverage, binned; group sizes are printed on the axis. **(b)** Observed share by tertile of the food's total PubMed literature (L1). **(c)** The planned primary model: fitted probability across lay-source coverage with literature volume held at its 25th, 50th, and 75th percentiles, with 95% confidence bands. Panels (a) and (b) are marginal and cannot separate the two predictors, because widely covered foods also tend to carry large literatures; panel (c) does separate them — the curves are flat across coverage and widely spaced by volume. Error bars in (a) and (b) are Wilson score intervals.

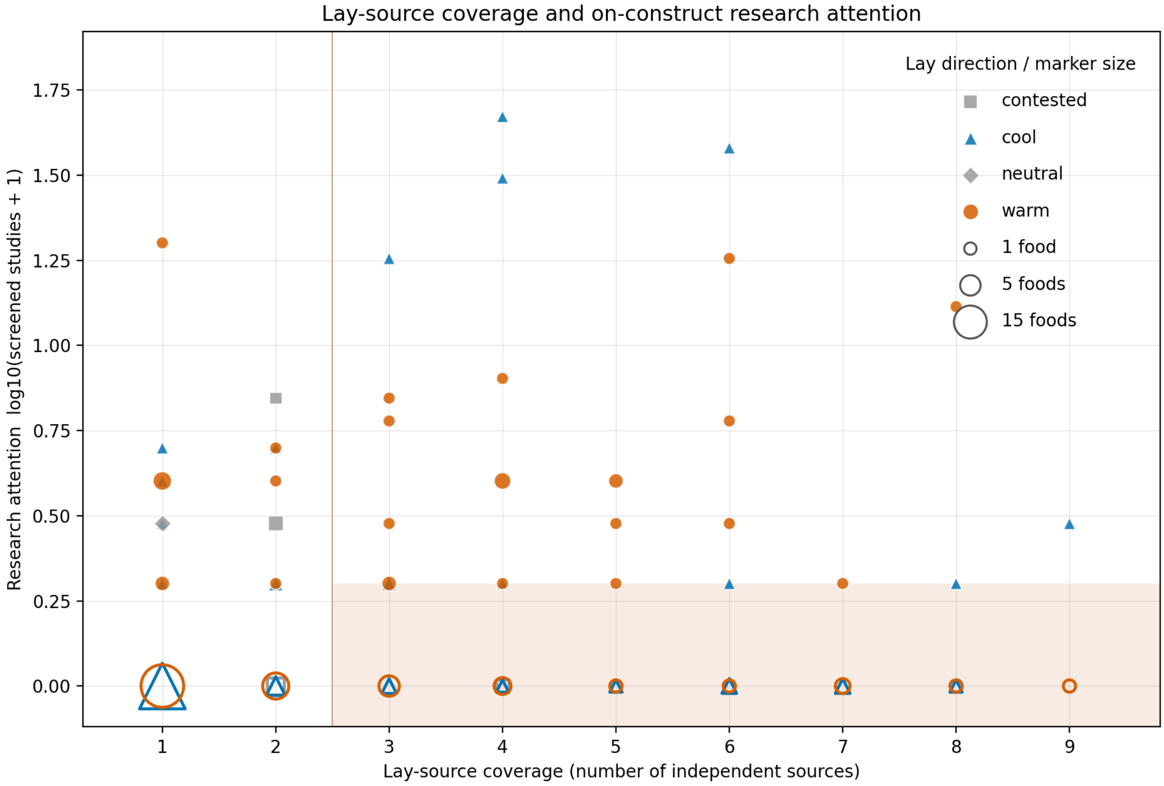

Figure 2. Lay-source coverage against on-construct research attention, per food (n = 146). x = number of independent Tier-1 sources assigning a warming or cooling direction; y = log10(on-construct studies + 1), so foods with no study sit on the baseline rather than being dropped by the log scale. Lay direction is encoded by both colour and marker shape (warming = vermillion circle, cooling = blue triangle, contested/neutral = grey square/diamond), so identity never rests on colour alone; the Okabe–Ito palette is colour-blind safe. Foods with no on-construct study are drawn hollow, and **marker area encodes how many foods share a position** — 96 of 146 foods sit on the baseline, which a plain scatter would render as a handful of points. The shaded region marks the core set (≥ 3 sources) with at most one study. Individual foods are named in Table 4 and in `plots/attention_gap_data.csv` rather than labelled in place, because leader lines from the baseline would cross studied foods higher up.